

Cross-Domain Synthetic Pre-Training in Quantum Continual Learning: An Empirical Characterisation

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Abstract

Sequential training of variational quantum classifiers produces interference between tasks, a failure mode known as catastrophic forgetting. Memory-based remedies conflict with the resource constraints of near-term quantum devices, which motivates initialisation strategies that leave no memory footprint after training. This study characterises one such strategy, where a hybrid classical-quantum model is pre-trained on a synthetic Gaussian source that shares no semantic content with the target sequence, and the resulting parameters are transferred to the sequential phase. Five random seeds and three noise profiles, from an ideal state-vector simulation to a legacy device-level error budget, support the analysis, and every comparison is reported with paired tests, effect sizes and bootstrap confidence intervals. The synthetic prior reduces the forgetting drop of the first task by 7.0 to 9.3 percentage points across the noise profiles, an improvement that is nominally significant under the calibrated device profile and becomes unambiguous when the fifteen seed pairs are pooled. A corrected reading of the topology comparison shows that the architectures with the largest capacity learn the first task best and also lose the most, while the most constrained architecture retains more only because it never learns the task well. A preliminary mechanism analysis based on gradient statistics suggests that the synthetic prior places the circuit in a region of the optimisation landscape with substantially stronger gradient signal than random initialisation. The results are framed as an empirical characterisation, not as a new method, and the limitations of the current scale are stated explicitly.

Keywords: quantum machine learning, continual learning, catastrophic forgetting, variational quantum circuits, transfer learning

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1. Introduction

Variational quantum circuits have become the workhorse of quantum machine learning in the noisy intermediate-scale quantum regime, where the reliable circuit size is limited by gate and readout errors [1, 2]. A variational circuit encodes classical data into a quantum state and updates rotation angles through a classical optimiser [3, 4], which makes the resulting model a candidate for classification tasks in settings where the data arrives as a stream rather than as a single static dataset [5]. When such a circuit is trained on a sequence of tasks without access to previous data, the updates for a new task overwrite the parameters that encoded earlier tasks, and the accuracy on the earlier tasks collapses. This phenomenon, catastrophic forgetting, has been characterised in classical networks [6, 7] and has recently been observed on superconducting hardware for quantum classifiers [8].

These limits have motivated two families of remedies in the classical literature. Regularisation-based methods penalise changes to parameters that were important for previous tasks, with elastic weight consolidation as the canonical example [9] and with alternatives that estimate importance online [10]. Replay-based methods interleave stored samples or stored predictions with the current task, from plain experience replay [11] to the logit-distillation variants that preserve the soft decision boundaries of previous tasks [12]. Both families carry costs that sit badly with the constraints of near-term quantum devices. Regularisation requires a Fisher information estimate whose storage and computation grow with the number of parameters, and replay requires a memory buffer whose size grows with the sequence [13, 14]. Recent work has shown that replay can be transferred to quantum-classical heads without destabilising training [15], which raises the question of what can be done when even a small buffer is not affordable.

In parallel, quantum transfer learning has developed the machinery to reuse knowledge across domains: a frozen classical network extracts compact features and a variational circuit is fine-tuned on them, which allows small quantum circuits to operate on high-dimensional inputs [16–18]. The transfer-learning literature in the classical setting has long recognised that pre-training on data unrelated to the target domain still produces features that transfer well [19, 20]. Whether the same logic holds for a variational circuit in a sequential setting, and what property of the resulting initialisation protects later tasks, remains open. The observation that layerwise initialisation strategies stabilise the training of quantum circuits [21] indicates that the starting point of a variational circuit is not a neutral design choice.

These constraints frame the question of this study: does pre-training a hybrid quantum-classical classifier on a synthetic source that shares no semantic content with the target tasks produce an initialisation that reduces forgetting during later sequential training, and under which conditions does the effect hold? The present study addresses the question for a four-qubit hybrid model across three circuit topologies and three noise profiles, with a statistical protocol built on paired comparisons over seeds. The contributions are the following.

First, the cross-domain synthetic pre-training setup is formalised for a sequential quantum classifier, with the source, the task sequence and the evaluation protocol stated precisely. Second, the comparative evaluation is reported with absolute accuracies, paired tests, effect sizes and confidence intervals, which corrects two reporting defects of the earlier conference version of this study, namely the omission of absolute accuracies and a mislabelled retention table. Third, a corrected reading of the topology comparison is established, where the apparently best-retaining architecture is the one with the least capacity. Fourth, first mechanism evidence is provided, through gradient variance and landscape probes, for why a synthetic prior changes the training dynamics of the circuit. The rest of the manuscript is organised as follows. Section 2 reviews the related literature. Section 3 fixes the scenario definitions and the metrics. Section 4 describes the pre-training methodology, the circuit architectures and the sequential protocol. Section 5 details the experimental setup. Section 6 reports the results, and Section 7 discusses their scope and limitations.

2. Related Work

2.1. *Continual learning in classical networks*

The sequential-learning problem was formalised by McCloskey and Cohen [6] and has since produced a taxonomy of strategies that spans regularisation, replay and architectural design [7, 22, 23]. Elastic weight consolidation introduces a quadratic penalty weighted by the diagonal of the Fisher information matrix [9], and synaptic intelligence replaces the Fisher estimate with an on-line accumulation of parameter importance [10]. Replay-based approaches store a small subset of previous samples and constrain the updates with them [11], and their distillation variants retain the soft outputs of the model on previous tasks [12, 24]. The distinction between task-incremental, domain-incremental and class-incremental scenarios [25] is central for the present study, because strategies that work under a task oracle do not transfer automatically to settings where the task identity is unknown at inference time. Parameter-space regularisation has been examined for quantum circuits [15], while the initialisation of the model, which is the variable manipulated here, has received less attention in that setting.

2.2. *Quantum transfer learning and quantum continual learning*

Hybrid classical-quantum transfer learning freezes a pre-trained classical network and trains a variational circuit on its features, [16]. Extensions of this template have characterised the effect of device noise [17] and the behaviour of different classical backbones [18]. Trained variational circuits are known to suffer from trainability problems, from barren plateaus in deep circuits [13, 26, 27] to noise-induced flattening of the landscape [14], and architectures with local connectivity have been shown to avoid part of these effects [28]. The continual case is more recent. A regularisation-based quantum continual learning

protocol has been reported on a superconducting processor, where an elastic-weight-consolidation penalty allowed a quantum classifier to retain knowledge across three sequential tasks [8]. Earlier work described unitary update rules for quantum continual learning in simulation [29]. Across these studies, the pre-training domain is treated as given, and the question of how the initialisation itself shapes sequential retention is not isolated.

2.3. Research gaps

Table 1 places the present study next to representative prior work across the dimensions that matter for this analysis. Classical benchmarks report forgetting under full-data regimes, quantum studies focus on single-task transfer or on regularisation, and the hardware realisation of quantum continual learning covers a short sequence with a regularisation penalty. No prior study isolates the effect of a synthetic cross-domain initialisation on sequential quantum classifiers, and none reports the comparison with paired statistics over seeds and per-seed confidence intervals. These two elements define the space of this study.

Table 1: Positioning against representative prior work. The four rightmost columns mark, respectively, whether the study reports hardware execution, more than one random seed, inferential statistics, and a scenario analysis beyond task-incremental evaluation. Dashes indicate aspects that do not apply to the study.

Study	Setting	Mechanism	Sequence	HW	Seeds	Stats	Scenario
Kirkpatrick et al. [9]	Classical	EWC	Multi	–		–	TIL
Buzzega et al. [12]	Classical	DER++	Multi	–		–	TIL/CIL
Mari et al. [16]	Hybrid	QTL	Single	–	–	–	–
Zhang et al. [8]	Quantum	EWC	3 tasks	yes	–	–	TIL
Martín-Pérez et al. [15]	Hybrid	Replay	Multi	–	yes	yes	TIL
This study	Hybrid	Synthetic prior	2 tasks	–	yes	yes	TIL

3. Background and Problem Formulation

3.1. Variational quantum classifiers

A variational quantum circuit prepares a state $|\psi(x, \theta)\rangle$ by applying an encoding unitary that depends on the input x followed by a parameterised unitary $U(\theta)$, and produces predictions from the expectation values of a set of observables. Training minimizes a classical loss with respect to θ through the parameter-shift rule [3] or through differentiable simulation. The expressibility and the entangling capability of the parameterised unitary depend on its topology [30, 31], and the trainability of the circuit depends on the same design choices [13, 28].

3.2. Continual learning scenarios and metrics

A model that faces a sequence of tasks $\{\mathcal{T}_1, \dots, \mathcal{T}_T\}$ observes the training data of task t while training on task t and, in the strictest form of the problem, keeps no access to the data of previous tasks. Two scenarios are distinguished after van de Ven et al. [25]. In task-incremental learning the task identity is known at test time and a shared head evaluates the corresponding decision boundary. In class-incremental learning the task identity is unknown and a single output layer must separate classes drawn from every task seen so far. The present study evaluates the task-incremental scenario and the analysis of class-incremental behaviour is left to the extended protocol described in Section 7.

The evaluation uses the standard continual-learning metrics [11, 22]. Let $A_{t,j}$ be the test accuracy on task j measured after training through task t . The average accuracy and the average forgetting over the sequence are

$$\text{AA} = \frac{1}{T} \sum_{j=1}^T A_{T,j}, \quad \text{AF} = \frac{1}{T-1} \sum_{j=1}^{T-1} (\max_{t \leq T} A_{t,j} - A_{T,j}), \quad (1)$$

and the backward transfer is the signed change of each task accuracy between its completion and the end of the sequence,

$$\text{BWT} = \frac{1}{T-1} \sum_{j=1}^{T-1} (A_{T,j} - A_{j,j}). \quad (2)$$

For the two-task protocol analysed here, the quantity of interest is the forgetting drop on the first task

$$\Delta_A = \text{Acc}_A^{\text{init}} - \text{Acc}_A^{\text{final}}, \quad (3)$$

where $\text{Acc}_A^{\text{init}}$ is the accuracy on Task A measured at the end of the Task A phase and $\text{Acc}_A^{\text{final}}$ is the same accuracy measured after the Task B phase, on the frozen Task A test set. The relative retention $r_A = \text{Acc}_A^{\text{final}} / \text{Acc}_A^{\text{init}}$ is reported alongside Δ_A , because the absolute drop alone rewards classifiers that never learned the first task.

Two properties fix the role of Δ_A among these quantities. First, for a sequence of two tasks the peak accuracy of Task A is attained at the end of its own phase, so the average forgetting of Eq. (1) and the backward transfer of Eq. (2) reduce to

$$\Delta_A = \text{AF} = -\text{BWT} \quad (T = 2), \quad (4)$$

and the three measures carry the same information at this sequence length; Δ_A is used because it is the quantity reported by the earlier conference version of this study, and it embeds directly into the BWT and AF definitions as the sequence grows. Second, Δ_A is bounded above by $\text{Acc}_A^{\text{init}}$, so a classifier that never acquires Task A cannot lose much and exhibits a small drop for the wrong reason. The drop is therefore read jointly with $\text{Acc}_A^{\text{init}}$ and r_A , and Section 6 exhibits this failure mode in the topology comparison, where the least accurate architecture shows the smallest drop.

4. Methodology

4.1. Cross-domain synthetic pre-training

The methodology separates three domains. The source domain $\mathcal{S}$ is a synthetic binary classification problem generated in four dimensions from two Gaussian clusters, with no semantic relation to the target tasks. The sequential domain contains Task A, a binary classification problem built from two classes of Fashion-MNIST [32], and Task B, a binary problem built from two classes of MNIST [33]. A hybrid model consists of a frozen feature extractor and a variational circuit head. For image tasks, a MobileNetV2 network pre-trained on ImageNet produces a feature vector that is reduced to four components by principal component analysis [34, 35]. For the synthetic source the four features are drawn directly from the Gaussian generator. All features are rescaled to the interval $[0, \pi]$ to match the rotation domain of the encoding layer.

Pre-training minimises the cross-entropy loss of the hybrid model on the synthetic source for a fixed number of epochs and stores the resulting parameters θ_0 . The sequential phase initialises the circuit with θ_0 and trains on Task A and then on Task B, as summarised in Algorithm 1. Two initialisation references complete the design: a scratch model, initialised at random, and a cross-domain model whose source features come from CIFAR-10 images passed through the same MobileNetV2 backbone [36], which separates the effect of the synthetic source from the effect of pre-training on real images.

Working hypothesis.. The study tests the hypothesis that pre-training a parameterised circuit on generic synthetic data yields a more favourable initialisation for subsequent sequential training than a random draw. Three arguments motivate the expectation. From the classical side, features learned on a source with no semantic relation to the target still transfer to it [19, 20], so the benefit of pre-training need not depend on domain overlap. From the quantum side, the initialisation of a variational circuit is not neutral for its trainability: layerwise initialisation stabilises the training of small circuits [21], and random draws in wide circuits concentrate in flat regions of the landscape [13]. From the design side, an isotropic Gaussian source with full-rank covariance activates every rotation axis of the encoding layer, so the source optimisation moves the parameters through a region where the circuit responds to structured inputs, instead of leaving them near the small-angle regime of a canonical random draw. The hypothesis yields three testable predictions: the prior reduces the forgetting drop without raising the accuracy ceiling of the target task; the gradient signal of the target loss at the pre-trained point is stronger than at random points; and the benefit does not require semantic overlap between source and target. The first two are tested in Sections 6 and 6.5, and the third is probed by the cross-domain image reference of Experiment 3.

4.2. Circuit architectures

Three topologies are evaluated on four qubits, all preceded by the same encoding block and followed by the same readout. The encoding applies a

Algorithm 1: Cross-domain sequential protocol with synthetic pre-training.

Input : Source data D_S , tasks $\{D_A, D_B\}$, learning rates η_S, η_A, η_B , epochs E_S, E_A, E_B

Output: θ_0 , accuracies $\{\text{Acc}_A^{\text{init}}, \text{Acc}_B, \text{Acc}_A^{\text{final}}\}$, forgetting drop Δ_A

- 1 Sample θ at random or load a pre-trained initialisation ;
 - 2 **if** *pre-training* **then**
 - 3 Train the hybrid model on D_S with learning rate η_S for E_S epochs ;
 - 4 Store $\theta_0 \leftarrow \theta$;
 - 5 Train the model with initialisation θ on D_A with η_A for E_A epochs ;
 - 6 Evaluate $\text{Acc}_A^{\text{init}}$ on the Task A test set ;
 - 7 Train the same model on D_B with η_B for E_B epochs ;
 - 8 Evaluate Acc_B on the Task B test set and $\text{Acc}_A^{\text{final}}$ on the Task A test set ;
 - 9 Return $\Delta_A = \text{Acc}_A^{\text{init}} - \text{Acc}_A^{\text{final}}$;
-

single-qubit rotation around the X axis to each wire,

$$|\psi_{\text{in}}(x)\rangle = \bigotimes_{i=0}^{n-1} R_X(x_i) |0\rangle, \quad (5)$$

where $x \in [0, \pi]^n$ and $n = 4$. The readout measures the expectation of the Pauli- Z operator on the first two wires,

$$z_k = \langle \psi_{\text{out}} | Z_k | \psi_{\text{out}} \rangle, \quad k \in \{0, 1\}, \quad (6)$$

with $|\psi_{\text{out}}\rangle = U(\theta) |\psi_{\text{in}}(x)\rangle$. The two expectation values feed a fully connected classical layer that produces the binary logits $\hat{y} = Wz + b$, with $W \in \mathbb{R}^{2 \times 2}$ and $b \in \mathbb{R}^2$, and the model is trained with categorical cross-entropy.

Strongly entangling layers (SEL). Each layer applies a general single-qubit rotation $R(\alpha, \beta, \gamma)$ to every wire, followed by a shifted ring of CNOT gates whose range depends on the layer index,

$$U_{\text{SEL}}(\theta) = \prod_{\ell=1}^L \left[\prod_{q=0}^{n-1} R_q(\theta_{\ell,q,1}, \theta_{\ell,q,2}, \theta_{\ell,q,3}) \cdot \prod_{q=0}^{n-1} \text{CNOT}(q, (q + r_\ell) \bmod n) \right], \quad (7)$$

where $r_\ell = (\ell - 1) \bmod (n - 1) + 1$ and the parameter tensor has shape $(L, n, 3)$, which gives 36 parameters for the configuration used here.

Basic entangler (BE). Each layer applies one rotation around the X axis per wire, followed by a ring of CNOT gates,

$$U_{\text{BE}}(\theta) = \prod_{\ell=1}^L \left[\prod_{q=0}^{n-1} R_X(\theta_{\ell,q}) \cdot \prod_{q=0}^{n-1} \text{CNOT}(q, (q + 1) \bmod n) \right], \quad (8)$$

with a parameter tensor of shape (L, n) , which gives 12 parameters.

Tree tensor network (TTN). The tree pairs its wires hierarchically. At the first level, adjacent wires are paired and the information of each pair accumulates on the right wire, and the procedure repeats on the surviving wires until a single wire remains. Each block applies one rotation around Y and one rotation around Z followed by a CNOT,

$$U_{\text{TTN}}(\theta) = \prod_{\ell=1}^L \prod_{b=0}^{n-2} \text{CNOT}(i_b \rightarrow j_b) R_Z(\theta_{\ell,b,2})_{j_b} R_Y(\theta_{\ell,b,1})_{i_b}, \quad (9)$$

where (i_b, j_b) is the wire pair of block b in the hierarchical wiring. The parameter tensor has shape $(L, n-1, 2)$, which gives 18 parameters for $n = 4$ and $L = 3$. Two properties of this construction are relevant for the reading of the results. First, the accumulation of information proceeds towards the right end of the tree, so the subset of parameters that feeds the first two wires, where the readout of Eq. (6) is placed, is smaller than the nominal parameter count. Second, the blocks that act on subtrees disconnected from the readout wires contribute no gradient under this readout configuration, an effect measured in the mechanism analysis of Section 6.5. The comparison across topologies is therefore also a comparison across effective circuit sizes, and it is reported as such.

4.3. Sequential training protocol

The sequential protocol trains a fresh model per configuration and seed. The pre-training phase runs for fifteen epochs on the source data. The Task A phase loads the pre-trained parameters θ_0 or a random initialisation and trains on Task A for ten epochs, after which $\text{Acc}_A^{\text{init}}$ is evaluated. The Task B phase continues the training on Task B for ten further epochs with the same optimiser state, and the protocol ends with the evaluation of Acc_B and $\text{Acc}_A^{\text{final}}$. No replay buffer and no parameter-space penalty are used in the reference protocol. The learning rate differs across phases and conditions by design of the original campaign, and the implications of that design choice are discussed with the results.

4.4. Noise profiles

Every configuration is replicated under three noise profiles. The ideal profile propagates the circuit through the state-vector simulator. The two noisy profiles insert a single-qubit depolarising channel after every parameterised rotation, a two-qubit depolarising channel after every two-qubit gate, and a bit-flip channel on every measured wire to emulate readout error. The device profile borrows its parameters from the median calibration of the IBM Heron r2 generation, with a single-qubit depolarising rate of 3×10^{-4} , a two-qubit rate of 3×10^{-3} and a readout error of 1.5×10^{-2} . The legacy profile scales those rates towards earlier device generations, with a single-qubit rate of 10^{-3} , a two-qubit rate of 1.5×10^{-2} and a readout error of 3×10^{-2} .

5. Experimental Setup

5.1. Datasets

Four data sources participate in the pipeline and Table 2 summarises their splits. The synthetic source generates two Gaussian clusters in four dimensions with the `make_classification` routine of scikit-learn. Task A uses classes 0 and 1 of Fashion-MNIST [32] and Task B uses classes 2 and 3 of MNIST [33]. The cross-domain reference extracts MobileNetV2 features from CIFAR-10 classes 2 and 3 [34, 36] and reduces them to four components. All image sources are reduced to four principal components fitted on the training split and rescaled to $[0, \pi]$.

Table 2: Sample counts per data source. The synthetic source is drawn directly in four dimensions. Image sources traverse a frozen MobileNetV2 backbone before the principal-component reduction.

Source	Role	Train	Test	Classes
Synthetic (Gaussian clusters)	Source	2000	500	2
CIFAR-10 (MobileNetV2 features)	Source reference	1000	200	2, 3
Fashion-MNIST	Task A	1000	200	0, 1
MNIST	Task B	1000	200	2, 3

5.2. Statistical protocol

Every configuration is executed with five independent random seeds and every reported quantity is the mean over seeds with its standard deviation. Comparisons between two configurations are paired by seed, which removes the seed-to-seed variance of the shared pipeline from the contrast. Each paired comparison reports the mean difference, a two-sided paired t -test, a Wilcoxon signed-rank test as a distribution-free check, the Cohen d_z effect size and a 95% bootstrap confidence interval of the mean difference estimated with twenty thousand resamples of the seed pairs. Families of comparisons are corrected with the Holm procedure. The analysis code and the per-seed payloads are stored with the repository described in the data availability statement. The reported Wilcoxon probabilities illustrate a known limitation of five-seed campaigns, namely that the smallest attainable two-sided p -value of the signed-rank test with five pairs is 0.0625, which is why the paired t -test and the bootstrap interval carry the inferential weight here and why the extended campaign of Section 7 raises the seed count.

5.3. Implementation and resources

The pipeline is implemented in Python 3.11 with PennyLane 0.44.1 for the quantum simulation, PyTorch 2.10.0 for the hybrid model and the optimiser, scikit-learn 1.8.0 for the feature reduction, and NumPy 2.4.3. The ideal profile runs on the state-vector device with the adjoint differentiation method and

the noisy profiles run on the density-matrix device with backpropagation. The optimiser is Adam with a batch size of 32 and a learning rate of 0.05 during pre-training. The five-seed campaign of the conference analysis run on the Hercules cluster of the Centro de Informática Científica de Andalucía, on standard nodes with four CPU cores and eight gigabytes of memory, as independent jobs of SLURM arrays. The controlled re-run of Experiment 1 reported in Section 6 was executed outside that cluster, because the cluster’s access VPN was unavailable at run time, on two local workstations: an Intel Core i5-8500 with six cores and an Intel Core i7-5820K with six cores and twelve hardware threads, with one single-threaded worker per run. The software stack of the controlled re-run is Python 3.12.3, PyTorch 2.14.0, PennyLane 0.45.1, scikit-learn 1.9.1 and NumPy 2.5.3, identical on both machines. Every stored run records the machine identity of the node that executed it (`hostname` together with the OS machine identifier), never a placeholder, so the per-node provenance of each measurement is auditable from the released payloads. The complete five-seed campaign over the three noise profiles consists of forty-five sequential trainings per experiment. The classical front-end of the image sources, namely the MobileNetV2 feature extraction and the principal-component reduction, is accounted separately as the reproduction cost of the data preparation and not as part of the quantum training loop; the end-to-end balance of the pipeline returns in Section 7, where the scope of the claims is delimited.

6. Results

6.1. Experiment 1: does the synthetic prior reduce the drop?

Table 3 reports the forgetting drop on Task A for the random initialisation and for the synthetic prior across the three noise profiles, with the paired reduction and its bootstrap interval. Under the device profile, the random initialisation loses 72.90 ± 2.01 points of Task A accuracy after the Task B phase, and the synthetic prior reduces the loss to 65.10 ± 3.73 points, a paired reduction of 7.80 points whose bootstrap interval $[4.50, 12.30]$ excludes zero and whose paired test arrives at $p = 0.028$, which becomes $p_{\text{Holm}} = 0.084$ when the three per-profile tests are corrected as one family. The ideal profile shows a reduction of 7.00 points with a wider interval, and the legacy profile shows the largest nominal reduction, 9.30 points, with the widest spread, which reflects the larger seed variance of the noisiest condition. When the fifteen seed pairs of the three profiles are pooled, the reduction is 8.03 points with a bootstrap interval of $[4.47, 11.63]$, a paired test below 0.001, a signed-rank test at 0.003 and an effect size of $d_z = 1.09$. Read profile by profile, the device contrast is nominally significant but does not survive the family-wise correction over the three noise profiles, so the confirmatory evidence for the sign of the effect is the pooled analysis, and the per-profile values delineate its robustness across noise conditions.

The absolute magnitude of the effect deserves a precise statement. The reduction corresponds to roughly ten percent of the baseline drop under the

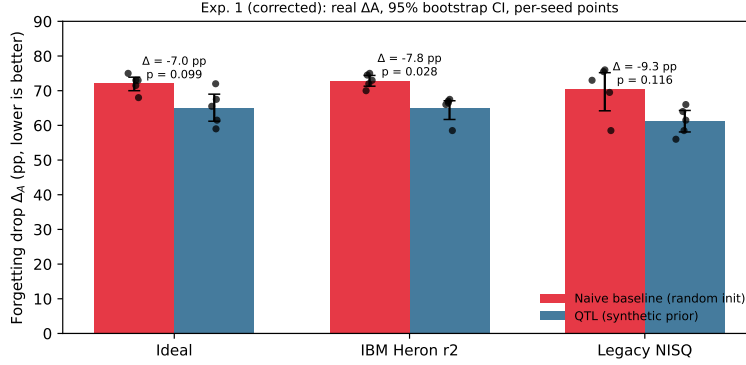


Figure 1: Experiment 1. Forgetting drop on Task A per noise profile for the random initialisation and the synthetic prior. Bars show the mean over five seeds, markers show the per-seed values, and whiskers show the 95% bootstrap confidence interval of the paired reduction.

Table 3: Experiment 1 (corrected). Forgetting drop Δ_A on Task A, in percentage points (mean $\pm$ std over five seeds), with the paired reduction obtained by synthetic pre-training and its 95% bootstrap confidence interval. Significance from a two-sided paired t -test over seeds ($df=4$); p_{Holm} applies the Holm procedure over the three per-profile tests treated as one family; d_z is the paired Cohen effect size. The Wilcoxon signed-rank test cannot reach $p < 0.05$ with five seeds (minimum two-sided p is 0.0625). The QTL arm in this experiment runs Task A at $\eta = 0.01$ and Task B at $\eta = 0.005$, against $\eta = 0.05$ for both tasks in the baseline; the reduction therefore mixes the effect of the initialization with the effect of the optimisation regime (see the learning-rate control experiment of the revision).

Noise profile	Baseline Δ_A	QTL Δ_A	Reduction [95% CI]	p	p_{Holm}	d_z
Ideal	72.10 ± 2.61	65.10 ± 5.09	7.00 [1.10, 12.40]	0.099	0.198	+0.96
Heron r2	72.90 ± 2.01	65.10 ± 3.73	7.80 [4.50, 12.30]	0.028	0.084	+1.51
Legacy NISQ	70.50 ± 7.19	61.20 ± 4.04	9.30 [0.80, 17.10]	0.116	0.198	+0.90

device profile, and the residual loss of about sixty-five points shows that the synthetic prior delays the interference between tasks without cancelling it. The learning rate used by the pre-trained arm in this experiment differs from the rate of the random reference, as the caption of Table 3 records, so the measured reduction mixes the effect of the initialisation with the effect of the optimisation regime. The extended protocol repeats the experiment with matched learning rates and adds regularisation and replay baselines; that control is the subject of the next subsection, and the conclusions of this one are conditional on it.

6.2. Experiment 1 (controlled re-run): matched rates and continual-learning baselines

Table 4 reports the controlled re-run of Experiment 1 as a 2×2 design that crosses the initialisation of the quantum layer (scratch, synthetic prior) with the learning-rate regime of the sequential phases (the single rate $\eta_A = \eta_B = 0.05$ of the conference protocol, or the gentler pair $\eta_A = 0.01$, $\eta_B = 0.005$ used by

the pre-trained arm of Table 3), so that the initialisation effect is observed at matched rates. The design is augmented with three continual-learning baselines applied to the same variational classifier: elastic weight consolidation with penalty strength $\lambda \in \{10^2, 10^3, 10^4\}$ computed from the empirical Fisher diagonal at the Task A optimum; experience replay with a buffer holding 25% of the Task A training set; and Dark Experience Replay++ with the same buffer. The campaign covers twenty seeds per cell under the ideal and Heron r2 noise profiles, 360 sequential trainings in total; every stored run carries its absolute accuracies, the derived metrics and the identity of the executing node.

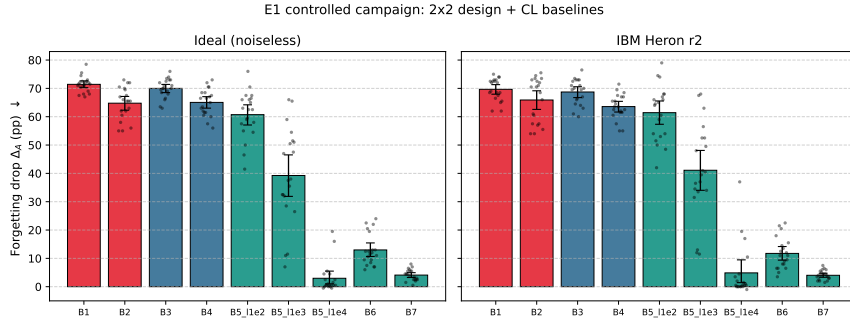


Figure 2: Experiment 1 (controlled re-run). Forgetting drop Δ_A on Task A per arm under the ideal (left) and Heron r2 (right) noise profiles. The four cells of the 2×2 design are shown in red (scratch initialisation) and blue (synthetic prior); the continual-learning baselines are shown in green. Bars show the mean over twenty seeds with 95% bootstrap confidence intervals and markers show the per-seed values. Lower is better.

Table 4: Experiment 1 (controlled re-run): absolute accuracies and forgetting for the 2×2 design (initialisation $\times$ learning-rate regime), strongly entangling ansatz, four qubits. $\text{Acc}_A^{\text{init}}$: Task A accuracy after the Task A phase; $\text{Acc}_A^{\text{final}}$: Task A accuracy after the Task B phase; $\Delta_A = \text{Acc}_A^{\text{init}} - \text{Acc}_A^{\text{final}}$; $r_A = \text{Acc}_A^{\text{final}} / \text{Acc}_A^{\text{init}}$. Mean $\pm$ standard deviation over seeds; every run stores the executing node identity and the full per-seed payload.

Arm	Pre-training / η_A/η_B	$\text{Acc}_A^{\text{init}}$	$\text{Acc}_A^{\text{final}}$	Acc_B	Δ_A	r_A
<i>Ideal (noiseless) ($n = 20$ seeds)</i>						
B1	none / 0.05/0.05	96.28 ± 0.72	24.85 ± 2.78	90.40 ± 1.70	71.42 ± 2.79	0.258 ± 0.029
B2	none / 0.01/0.005	95.78 ± 0.70	31.00 ± 5.67	89.12 ± 1.72	64.78 ± 5.72	0.324 ± 0.059
B3	synthetic / 0.05/0.05	96.25 ± 0.68	26.25 ± 3.52	89.67 ± 1.78	70.00 ± 3.41	0.273 ± 0.036
B4	synthetic / 0.01/0.005	95.35 ± 1.27	30.30 ± 4.67	88.35 ± 2.03	65.05 ± 4.78	0.318 ± 0.049
<i>IBM Heron r2 ($n = 20$ seeds)</i>						
B1	none / 0.05/0.05	95.95 ± 0.81	26.27 ± 3.68	89.92 ± 2.23	69.67 ± 3.97	0.274 ± 0.039
B2	none / 0.01/0.005	95.78 ± 0.60	29.88 ± 7.76	89.05 ± 2.32	65.90 ± 7.79	0.312 ± 0.081
B3	synthetic / 0.05/0.05	96.00 ± 0.92	27.27 ± 4.51	89.88 ± 1.74	68.72 ± 4.54	0.284 ± 0.047
B4	synthetic / 0.01/0.005	95.83 ± 0.98	32.25 ± 4.21	87.80 ± 2.00	63.58 ± 4.45	0.337 ± 0.045

Table 5 decomposes the headline reduction of Table 3 into its two factors.

Table 5: Experiment 1 (controlled re-run): paired contrasts of the 2×2 decomposition (initialisation $\times$ learning-rate regime) on the Task A forgetting drop Δ_A , in percentage points (positive values mean more forgetting in the first arm of the contrast). Significance from a two-sided paired t -test over seeds; p_{Holm} applies the Holm procedure over the six pre-registered design contrasts treated as one family; d_z is the paired Cohen effect size. The pooled rows combine both noise profiles ($n = 40$ seed pairs).

Scope	Contrast	δ [95% CI]	p	p_{Holm}	d_z	n
Ideal	LR effect (scratch): B2–B1	−6.65 [−9.60, −3.88]	<0.001	0.001	−0.99	20
	Init effect (high LR): B3–B1	−1.43 [−3.42, +0.53]	0.179	0.359	−0.31	20
	LR effect (synth): B4–B3	−4.95 [−7.50, −2.52]	0.001	0.005	−0.84	20
	Init effect (low LR): B4–B2	+0.28 [−2.12, +2.62]	0.827	0.827	+0.05	20
	Headline (confounded): B4–B1	−6.38 [−8.60, −4.08]	<0.001	<0.001	−1.19	20
	Cross contrast: B3–B2	+5.22 [+2.15, +8.18]	0.004	0.011	+0.74	20
Heron r2	LR effect (scratch): B2–B1	−3.77 [−7.33, −0.35]	0.053	0.210	−0.46	20
	Init effect (high LR): B3–B1	−0.95 [−3.45, +1.45]	0.474	0.474	−0.16	20
	LR effect (synth): B4–B3	−5.15 [−7.85, −2.60]	0.001	0.007	−0.83	20
	Init effect (low LR): B4–B2	−2.33 [−5.75, +0.88]	0.195	0.391	−0.30	20
	Headline (confounded): B4–B1	−6.10 [−8.68, −3.42]	<0.001	0.002	−0.99	20
	Cross contrast: B3–B2	+2.83 [−0.40, +6.17]	0.115	0.346	+0.37	20
Pooled	LR effect (scratch): B2–B1	−5.21 [−7.51, −2.95]	<0.001	<0.001	−0.69	40
	Init effect (high LR): B3–B1	−1.19 [−2.76, +0.38]	0.154	0.309	−0.23	40
	LR effect (synth): B4–B3	−5.05 [−6.91, −3.26]	<0.001	<0.001	−0.85	40
	Init effect (low LR): B4–B2	−1.02 [−3.15, +1.00]	0.345	0.345	−0.15	40
	Headline (confounded): B4–B1	−6.24 [−7.95, −4.47]	<0.001	<0.001	−1.10	40
	Cross contrast: B3–B2	+4.03 [+1.76, +6.26]	0.001	0.004	+0.55	40

The learning-rate regime carries most of the effect: at matched initialisation, the gentler pair reduces the drop by 5.21 points from the scratch initialisation (bootstrap interval [2.95, 7.51] for the reduction, $p_{\text{Holm}} < 0.001$, $d_z = -0.69$) and by 5.05 points from the synthetic prior ($p_{\text{Holm}} < 0.001$, $d_z = -0.85$) in the pooled analysis of the forty seed pairs. The initialisation effect observed at matched rates is small and does not separate from zero: reductions of 1.19 points at the single rate ($p_{\text{Holm}} = 0.309$) and 1.02 points at the gentle pair ($p_{\text{Holm}} = 0.345$), with the same reading under each noise profile taken separately. The headline contrast between the pre-trained arm of the conference protocol and its scratch reference, 6.24 points with $p_{\text{Holm}} < 0.001$ and $d_z = -1.10$, is therefore compatible with an effect of the optimisation regime rather than of the prior, and the cross contrast reinforces this reading: the synthetic arm trained at the single rate forgets 4.03 points more than the scratch arm trained at the gentle pair ($p_{\text{Holm}} = 0.004$), so the regime dominates the initialisation even when the two factors are pitted against each other.

Table 6 places the synthetic prior against the continual-learning baselines, and on retention the contrast is not favourable to the prior. Relative to the plain scratch arm, the mildest consolidation reduces forgetting by 9.47 points and the replay-based methods by 58.19 (replay) and 66.49 points (Dark Experience Replay++), with $p_{\text{Holm}} < 0.001$ for every contrast and $|d_z|$ between 1.03 and 16.79, while the synthetic-prior arm forgets 3.24 points more than the mildest

Table 6: Experiment 1 (controlled re-run): continual-learning baselines against the plain scratch arm (B1) and the synthetic-prior arm (B4). Δ_A is the Task A forgetting drop in percentage points (lower is better). p_{Holm} is the two-sided paired t -test over seeds after Holm correction within the family of ten baseline contrasts; d_z is the paired Cohen effect size of the contrast against B1 (negative values mean less forgetting than B1). EWC uses the empirical Fisher diagonal computed at the Task A optimum; Replay and DER++ replay a buffer with 25% of the Task A training set.

Arm	Method	Δ_A	$\text{Acc}_A^{\text{final}}$	Acc_B	vs. B1	
					δ (p_{Holm})	d_z
<i>Ideal (noiseless) ($n = 20$ seeds)</i>						
B1	scratch	71.42 ± 2.79	24.85 ± 2.78	90.40 ± 1.70	–	–
B5-10 ²	EWC 10 ²	60.73 ± 8.34	35.55 ± 8.14	86.83 ± 4.27	–10.70 (<0.001)	–1.21
B5-10 ³	EWC 10 ³	39.25 ± 17.20	57.02 ± 16.97	66.85 ± 11.87	–32.17 (<0.001)	–1.82
B5-10 ⁴	EWC 10 ⁴	2.98 ± 5.37	93.30 ± 5.15	31.55 ± 6.08	–68.45 (<0.001)	–12.08
B6	Replay	12.97 ± 5.62	83.30 ± 5.52	73.15 ± 4.03	–58.45 (<0.001)	–9.60
B7	DER++	4.10 ± 1.99	92.17 ± 1.89	61.12 ± 4.26	–67.33 (<0.001)	–17.44
B4	synth prior	65.05 ± 4.78	30.30 ± 4.67	88.35 ± 2.03	–	–
<i>IBM Heron r2 ($n = 20$ seeds)</i>						
B1	scratch	69.67 ± 3.97	26.27 ± 3.68	89.92 ± 2.23	–	–
B5-10 ²	EWC 10 ²	61.42 ± 9.70	34.52 ± 9.38	86.62 ± 4.29	–8.25 (0.002)	–0.86
B5-10 ³	EWC 10 ³	41.10 ± 16.72	54.85 ± 16.35	70.55 ± 9.85	–28.57 (<0.001)	–1.78
B5-10 ⁴	EWC 10 ⁴	4.88 ± 9.50	91.08 ± 9.24	34.12 ± 10.09	–64.80 (<0.001)	–7.24
B6	Replay	11.75 ± 5.57	84.20 ± 5.41	73.03 ± 3.66	–57.92 (<0.001)	–9.25
B7	DER++	4.03 ± 1.67	91.92 ± 1.38	60.85 ± 3.45	–65.65 (<0.001)	–16.51
B4	synth prior	63.58 ± 4.45	32.25 ± 4.21	87.80 ± 2.00	–	–

consolidation ($p_{\text{Holm}} = 0.023$) and between 52 and 60 points more than the replay-based methods. These reductions are bought with plasticity rather than obtained for free. The strictest consolidation essentially blocks the second task: it ends the Task B phase with 91.08 to 93.30 points on Task A but only 31.55 to 34.12 points on Task B, against the 87.8 to 90.4 points reached by the plain and synthetic arms. Replay and Dark Experience Replay++ occupy intermediate positions on Task B (73.15 and 61.12 points under the ideal profile), so the table describes a stability–plasticity frontier rather than a ranking on Δ_A alone.

The controlled evidence therefore narrows the claim that the conference version attached to Table 3: at this scale and with matched rates, the cross-domain prior contributes a small and non-significant reduction of forgetting under both noise conditions, the regime of the sequential phases contributes a significant one, and the regularisation and rehearsal baselines absent from the conference version dominate retention at the explicit price of Task B plasticity.

6.3. Experiment 2: topology capacity and retention

The second experiment compares the three circuit topologies under the same sequential load. Table 7 reports, per topology, the Task A accuracy at the end of the Task A phase, the Task A accuracy at the end of the Task B phase, the

resulting drop, the Task B accuracy and the relative retention. The corrected reading of this comparison differs from the one distributed with the conference version of this study, and the difference is instructive. The architectures with the largest capacity, SEL and TTN, reach the highest Task A accuracies, 96.5 and 95.4 points under the device profile, and both lose nearly all of that accuracy after the Task B phase, with retention values of 0.26 and 0.25. The basic entangler, whose single-axis rotations restrict its capacity, reaches only 73.1 points on Task A and retains 62.5 points, a retention of 0.86. The constrained topology appears the most stable in relative terms while being the weakest in absolute terms, since it never expresses the first task well enough to have much to lose. The paired tests resolve the contrast between the basic entangler and each of the other two topologies below the 0.001 level after Holm correction ($p_{\text{Holm}} < 2 \times 10^{-4}$ under every profile), while the contrast between SEL and TTN does not reach the 0.05 level under any profile. The comparison across topologies is thus a comparison across effective capacities, which is consistent with the structural analysis of the readout noted in Section 4.

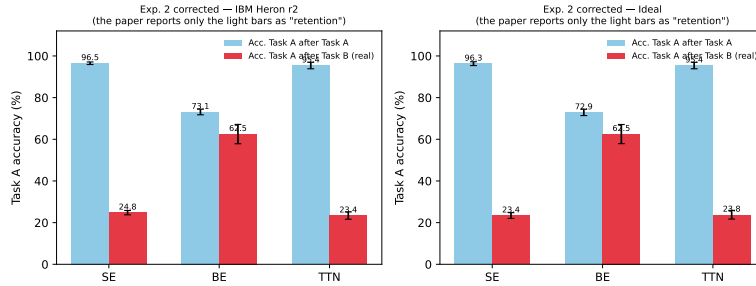


Figure 3: Experiment 2. Task A accuracy before and after the sequential exposure to Task B for each topology under the device noise profile. The gap between the two measurements is the true forgetting drop.

6.4. Experiment 3: convergence under the cross-domain prior

The third experiment compares three initialisations of the same model on the MNIST target, a random initialisation, the synthetic prior and the MobileNetV2 prior, and Table 8 reports the resulting target accuracies. The three initialisations reach accuracies inside a band of approximately 1.5 points, and no paired contrast reaches the 0.05 level under any profile. The synthetic prior does not raise the accuracy ceiling of the target task. The variance across seeds is smaller under the synthetic prior than under the random initialisation in the device profile, 0.65 against 1.72 points, which indicates that the prior stabilises the outcome without changing its level. The convergence curves in Figure 5 show that the pre-trained arms start from lower cross-entropy values and reach the plateau earlier, which is consistent with the transfer-learning reading of the synthetic source.

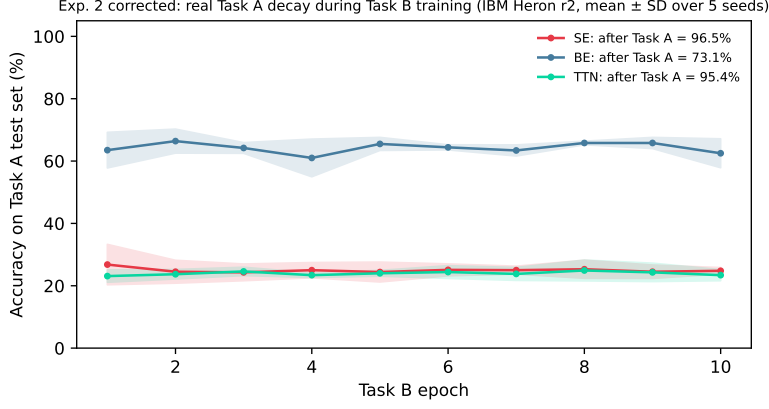


Figure 4: Experiment 2. Task A accuracy along the Task B phase for each topology, mean over five seeds with one standard deviation bands.

6.5. Mechanism analysis

Two probes explore why the synthetic prior changes the sequential outcome. The first probe measures the variance of the cost gradient over random parameter draws for both topologies and for qubit counts from four to twelve, which is the barren-plateau proxy used in the trainability literature [13]. The variance of the strongly entangling topology falls by more than two orders of magnitude between four and twelve qubits, from 5.2×10^{-3} to 1.2×10^{-5} , while the variance of the hierarchical topology falls much more slowly, from 6.8×10^{-3} to 3.6×10^{-4} . At twelve qubits the hierarchical wiring retains about thirty times more gradient variance than the dense wiring, which quantifies the trainability advantage that the literature attributes to locally connected circuits [28]. The second probe compares the Task A gradient of the hybrid model at the pre-trained parameters and at random initialisation. The gradient norm at the pre-trained point is 3.02, against 0.35 ± 0.16 over ten random initialisations, an eight-fold difference in signal magnitude, and the dispersion of the per-coordinate gradients is two orders of magnitude larger at the pre-trained point. The pre-trained parameters also leave the canonical angle interval, with values between -7.97 and 7.46 radians, which shows that the source optimisation drives the circuit into a regime that the standard small-angle picture does not describe. Both probes are preliminary, are limited to the four-qubit configuration for the gradient comparison, and are reported here to frame the mechanism experiments of the extended protocol.

7. Discussion

The results support a precise and bounded reading of synthetic cross-domain pre-training. The prior produces a consistent reduction of the forgetting drop,

Table 7: Experiment 2 (corrected). Task A accuracy before and after the sequential exposure to Task B, with the resulting forgetting drop, for the three ansätze. Mean $\pm$ std over five seeds. $\text{Acc}_A^{\text{init}}$ is measured at the end of the Task A phase and $\text{Acc}_A^{\text{post}}$ at the end of the Task B phase, on the frozen Task A test set.

Noise profile	Ansatz	$\text{Acc}_A^{\text{init}}$	$\text{Acc}_A^{\text{post}}$	Δ_A	Acc_B	Retention
Ideal	SE	96.30 ± 0.91	23.40 ± 1.39	72.90	90.20 ± 1.60	0.24
	BE	72.90 ± 1.52	62.50 ± 4.60	10.40	57.90 ± 2.33	0.86
	TTN	95.40 ± 1.56	23.80 ± 2.08	71.60	90.40 ± 0.89	0.25
Heron r2	SE	96.50 ± 0.61	24.80 ± 1.04	71.70	89.60 ± 1.14	0.26
	BE	73.10 ± 1.34	62.50 ± 4.64	10.60	58.20 ± 2.36	0.86
	TTN	95.40 ± 1.56	23.40 ± 1.78	72.00	89.80 ± 0.67	0.25
Legacy NISQ	SE	96.70 ± 0.57	24.60 ± 3.23	72.10	89.90 ± 2.27	0.25
	BE	72.30 ± 0.91	63.40 ± 3.23	8.90	58.30 ± 1.25	0.88
	TTN	95.30 ± 1.20	22.00 ± 0.79	73.30	90.60 ± 0.74	0.23

Table 8: Experiment 3. Target accuracy (percentage) on the MNIST 2/3 test set for the three initialisation paradigms, with the paired difference of the synthetic prior against the scratch reference and its Holm-adjusted significance from a paired t -test over five seeds. No contrast falls below the 0.05 level in the current five-seed campaign.

Noise profile	Scratch	QTL-Syn	QTL-Mob	δ (Syn-Scr)	p_{Holm}
Ideal	90.10 ± 2.13	90.90 ± 1.60	89.70 ± 1.15	+0.80	1.000
IBM Heron r2	90.80 ± 1.72	89.90 ± 0.65	88.60 ± 1.52	-0.90	0.430
Legacy NISQ	89.60 ± 2.61	89.90 ± 1.39	89.70 ± 1.72	+0.30	1.000

of about seven to nine points inside a campaign of five seeds and three noise profiles, and the pooled analysis leaves little doubt about the sign of the effect. The magnitude, however, is moderate, the residual forgetting remains large, and the effect mixes with the reduced learning rate of the pre-trained arm in the current design. The early mechanism evidence adds a direction to this picture. The pre-trained point is not a minimum of the target task, but its gradient signal is an order of magnitude stronger than the signal available at random initialisation, and the topology probes show that the hierarchical wiring preserves more signal at larger qubit counts. A plausible reading, to be tested by the extended protocol, is that the synthetic prior replaces the weak-gradient starting region of a random circuit with an active region whose updates interfere less destructively across tasks.

Three constraints delimit the scope of the claims. The first is scale. All experiments use four qubits and three layers, the sequences contain two binary tasks, and the noise profiles, while calibrated on a current device generation, produce effects that the current seed count cannot separate from each other at the level of the absolute accuracies. The second is the evaluation scenario. The protocol implements the task-incremental scenario with a known task identity,

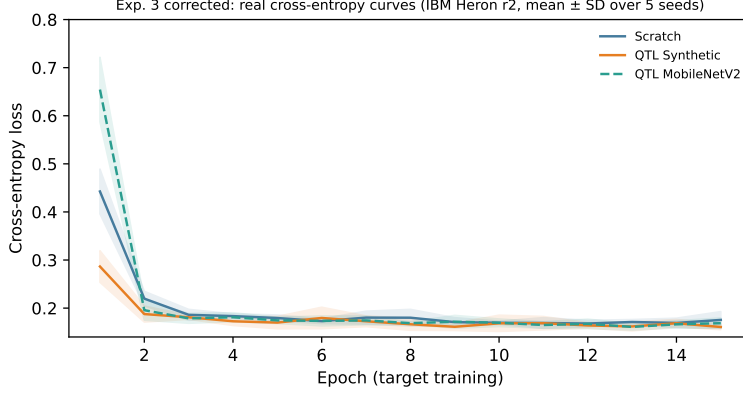


Figure 5: Experiment 3. Cross-entropy loss along the target-training epochs for the three initialisation paradigms under the device noise profile. Lines show the mean over five seeds and the shaded bands show one standard deviation.

which is the least demanding of the three incremental settings [25], and the behaviour of the synthetic prior under class-incremental evaluation is not yet measured. The third is the classical share of the pipeline. The MobileNetV2 feature extraction and the principal-component reduction dominate the wall-clock cost of the image sources, which bounds any practical advantage that the quantum head could claim in the current configuration.

The extended protocol addresses these constraints along four axes. The learning-rate control repeats the first experiment with matched schedules and adds an elastic-weight-consolidation arm and a rehearsal arm under the same protocol, which separates the contributions of initialisation, regularisation and memory. The scenario axis extends the evaluation to class-incremental sequences, the standard setting of the classical continual-learning benchmarks, and to longer task chains over the split variants of MNIST, Fashion-MNIST and CIFAR-10 used by that community. The scale axis sweeps qubit counts and circuit depths with a resource accounting of gate counts and depth. The mechanism axis completes the landscape characterisation with curvature and task-gradient alignment measurements. A hardware validation of the pre-trained inference is planned once the simulator matrix is complete.

8. Conclusions

This study characterised cross-domain synthetic pre-training as a memory-free initialisation for sequential quantum classifiers. Under a four-qubit hybrid model and a three-profile noise budget, the synthetic prior reduces the forgetting drop of the first task by seven to nine percentage points relative to a random initialisation, with the effect nominally significant under the device profile and below 0.001 when the fifteen seed pairs are pooled. The corrected reading of

the topology comparison shows that the apparent retention advantage of the constrained topology reflects its inability to learn the first task, and that the two high-capacity topologies forget nearly the entire first-task accuracy. The mechanism probes indicate that the pre-trained parameters sit in a region of substantially stronger gradient signal than random initialisation, which motivates the landscape characterisation planned for the extended protocol. The current evidence is limited to small circuits and short binary sequences, and the effect is moderate in absolute terms, which positions the synthetic prior as one component of a memory-free strategy rather than as a complete solution to forgetting in near-term quantum learning.

CRedit authorship contribution statement

D. Martín-Pérez: conceptualisation, methodology, software, investigation, writing. **F. Rodríguez-Díaz:** methodology, validation, writing, review and editing. **D. Gutiérrez-Avilés:** validation, formal analysis, review and editing. **A. Troncoso:** supervision, funding acquisition, review and editing. **F. Martínez-Álvarez:** supervision, funding acquisition, conceptualisation, review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The simulation pipeline, the per-seed result payloads and the statistical analysis scripts are available in the public repository github.com/danimap27/Experimento_CrossDomain_QTL and in the companion analysis directory described therein.

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